MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. $(x^3)'$ value
   - A. x^2
   - B. $3x^2$
   - C. $3x$
   - D. $\frac{x^4}{4}$
2. $(\frac{1}{x})'$ value
   - A. $-\frac{1}{x^2}$
   - B. $\frac{1}{x^2}$
   - C. $-\frac{1}{x}$
   - D. 1
3. $(uv)'$ value
   - A. $u'v'$
   - B. $u' + v'$
   - C. uv'
   - D. $u'v + uv'$
4. $f' > 0$ on an interval implies
   - A. f positive
   - B. f negative
   - C. f strictly increasing
   - D. f constant
5. $\frac{x^2+1}{x-1}$ is defined if
   - A. $x > 0$
   - B. $x \neq 1$
   - C. $x \neq -1$
   - D. $x \neq 0$

MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $x^3 - x$ is

   A. even

   B. constant

   C. odd

   D. neither even nor odd

2. $f\prime(a) = 0$ necessarily implies

   A. horizontal tangent

   B. maximum

   C. minimum

   D. f constant

3. Derivative of $\sqrt{x}$:

   A. $2\sqrt{x}$

   B. $\dfrac{1}{\sqrt{x}}$

   C. $\dfrac{\sqrt{x}}{2}$

   D. $\dfrac{1}{2\sqrt{x}}, x > 0$

4. Minimum of $x + \dfrac{1}{x}$ on $]0; +\infty[$:

   A. 0

   B. 2

   C. 1

   D. none

5. Newton's method for $x^2 - 2$ is undefined when $x_0 =$

   A. 1

   B. 2

   C. 0

   D. -1

True or false? Justify.

1. The derivative of a positive function is positive.

2. An interior extremum of a differentiable function gives a zero derivative.

3. The product of two increasing functions is increasing.

Personal review

I differentiate on the correct domain; I simplify without losing excluded values; I distinguish local and global extrema; I sometimes choose a factorisation rather than a derivative.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.